

Exploring Virtualization Efficiency, Security Enhancements, and Scripting for File and Operating Systems*

Matus ILLITHI

Student ID: 2325478

Robert Gordon University Aberdeen

School of Computing, Engineering and Technology

`m.illith@rgu.ac.uk`

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Abstract

... The gaming industry is one of the fastest-growing markets in the world. With rapid improvement in game development, there is no doubt that the computing power requirement keeps rising. With that comes new growing energy expenses. Virtualisation and performance have been in the eyes of researchers for years. This paper dives into the cost, energy efficiency and security of virtualisations for ACME Games. This includes examples of File Scripting and Operating System Scripting with adherence to proper coding ethics and security.

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1 Virtualisation Efficiency Benefits

1.1 Understanding virtualization

Virtualization in simple terms is sharing of physical resource between multiple virtual machines. You can think of it as computer within a computer, implemented in software. (toto je z jedneho articlu pridaj referencie). This is done by creating abstraction layer over physical hardware of host machine so multiple computers (Virtual Machines) can run on one hardware. They can also share resources such as network cards, CPU, memory, Physical storage and other. Virtual machine is virtual depiction of machines hardware which is maintained by virtual machine monitor (VMware Fusion, Oracle VM VirtualBox). (Abstraction diagram here[create/find on internet]) Concept of virtualization isn't new it has been here since 1960. But with introduction of X86 Architecture and platforms it now has been in sights of many companies because it provided simple seamless implementation without significant interruption to operations and low acquisition.

1.2 Ako spravit virtualizaciu

1.3 Benefits of Virtualization

The benefits of virtualization are considerable. Virtualization can help with security, scalability, cost cutting, efficient uses of resources and other. Non virtualized system can be inefficient when non virtualized resources aren't in use the hardware is sitting idle without any computing task ready to run. While virtualized system once resources become available hypervisor can allocate them accordingly to another task so the physical resources are always in use. With this comes another benefit of security and corruption prevention where in normal non virtualized system the security and backup layers are on the one specific machine and user. The virtualized system security depends on security of the whole system. This can lead to better and secure disaster prevention where we can back up each VM so in case of natural disaster there isn't any loss of data. In simple terms virtualization helps with growth scalability of a organization.

For ACME Games virtualization will not only help with cost cutting and security but also with:

1. Minimize Servers
2. Reduce Hardware
3. Quick Redeployment
4. Simpler Backups
5. Better Testing

1.4 Cost of virtualization and energy efficiency

This paper delves into total cost of ownership for virtualized systems compared to standard systems and energy efficiency of medium to large organizations. We will look in to direct and indirect costs of virtualization for organizations.

With real life examples of companies that went through restructuring of their servers to implement virtualization. First we need to define the major direct cost categories for the organization

1.4.1 Cost of virtualization

1. Hardware cost

Represented as expenses that are expected to be bought so virtualization can be implemented to the system. These include physical technologies that would allow to create new virtualized system

2. Software cost

Are expenses derived from buying and/or license software which allows access to virtual machines and all its possible functions.

3. Deployment costs

These costs are other expenses not directly associated with the concept of virtualization and virtualization itself. Costs like disruption to existing operation or training staff on new system.

These are direct cost of implementing virtualization they can be one time or recurring. These expenses represent general figure for medium to large firms. These can be cut down further with right licensing agreement and right negotiation with hardware providers.

Sample text with a reference. [1]

1.4.2 Energy efficiency

2 Security Issues

- 1.
- 2.
- 3.

3 File System Scripting

4 Operating System Scripting

5 Conclusions

References

- [1] Olli Helin Jukka Kommeri, Tapio Niemi. Energy efficiency of server virtualization. https://personales.upv.es/thinkmind/dl/conferences/energy/energy_2012/energy_2012_4_30_40149.pdf.